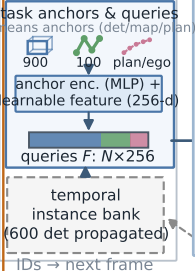
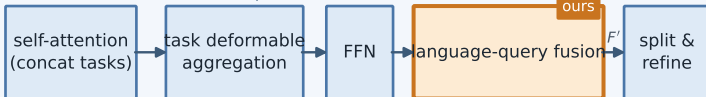


geometry pathway (HiP-AD base)

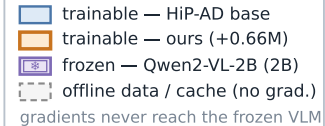


unified decoder layer

in: queries $F (N \times 256)$

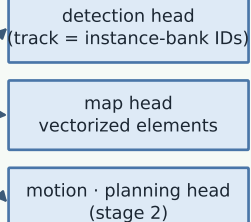


layers 2-6 also attend propagated temporal queries



updated queries
 $N \times 256$

task heads



language pathway (ours)

same 6 views, resized $200 \times 28 \times 28$ /cam

offline — once per sample, no VLM in training

